# Exponential Inequalities and Concentration of Measure

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**Notation and Conventions.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For a random variable $X$, we denote its expectation by $\mathbb{E}[X]$ and its variance by $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. The Moment Generating Function (MGF) of $X$ is $M_X(t) = \mathbb{E}[e^{tX}]$, defined for $t$ in a neighborhood of zero. The natural logarithm is denoted by log. We write $X \sim \mathcal{N}(\mu, \sigma^2)$ to indicate that $X$ follows a normal distribution with mean $\mu$ and variance $\sigma^2$.

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## 1 Fundamental Scalar Inequalities

The following analytic inequalities serve as the primitive tools from which probabilistic bounds are derived.

### 1.1 The Exponential-Cosh Inequality

For all $x \in \mathbb{R}$:

$$\frac{e^x + e^{-x}}{2} = \cosh(x) \leq e^{x^2/2}.$$

*Proof.* Consider the Maclaurin series of both functions:

$$\cosh(x) = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}, \qquad e^{x^2/2} = \sum_{k=0}^{\infty} \frac{x^{2k}}{2^k k!}.$$

It suffices to show that for all $k \geq 0$, $(2k)! \geq 2^k k!$. For $k = 0$, equality holds ($1 = 1$). For $k \geq 1$, observe that $(2k)! = (1 \cdot 2 \cdot \cdots \cdot k) \cdot ((k+1) \ldots (2k)) = k! \prod_{j=1}^{k} (k+j)$. Since $k + j \geq 2$ for all $j \geq 1$, we have $\prod_{j=1}^{k} (k+j) \geq 2^k$. Thus, $(2k)! \geq 2^k k!$, and the inequality follows term-by-term. □

### 1.2 Basic Exponential Bounds

For $x \in \mathbb{R}$:

1. $1 + x \leq e^x$ (with equality iff $x = 0$).
2. $\log(1 + x) \leq x$ for $x > -1$.
3. $e^x \geq 1 + x + \frac{x^2}{2}$ for $x \geq 0$.

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## 2 Moment Generating Functions and the Chernoff Method

The Chernoff method (or Chernoff-Cramér bound) is the standard technique for bounding tail probabilities using the MGF.

### 2.1 The Chernoff Bound

Let $X$ be a random variable and $a \in \mathbb{R}$. For any $t > 0$,

$$\mathbb{P}(X \geq a) = \mathbb{P}(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = \frac{M_X(t)}{e^{ta}}.$$

Similarly, for $t < 0$, $\mathbb{P}(X \leq a) \leq \frac{M_X(t)}{e^{ta}}$.

### 2.2 MGFs of Standard Distributions

- **Rademacher** ($X \in \{-1, 1\}$ equiprobably): $M_X(t) = \frac{e^t + e^{-t}}{2} = \cosh(t) \leq e^{t^2/2}$.
- **Normal** $\mathcal{N}(\mu, \sigma^2)$: $M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$.
- **Bernoulli** ($p$): $M_X(t) = 1 - p + pe^t$.

### 2.3 Hoeffding's Lemma

Let $X$ be a random variable with $\mathbb{E}[X] = \mu$ and $\mathbb{P}(a \leq X \leq b) = 1$. Then for all $t \in \mathbb{R}$:

$$\mathbb{E}[e^{t(X-\mu)}] \leq \exp\left(\frac{t^2(b-a)^2}{8}\right).$$

*Proof.* Without loss of generality, assume $\mu = 0$ (otherwise replace $X$ with $X - \mu$). Since $e^{tx}$ is convex, the graph of $e^{tx}$ on $[a, b]$ lies below the secant line connecting $(a, e^{ta})$ and $(b, e^{tb})$. Thus,

$$e^{tx} \leq \frac{b - x}{b - a}e^{ta} + \frac{x - a}{b - a}e^{tb}.$$

Taking expectations and using $\mathbb{E}[X] = 0$,

$$\mathbb{E}[e^{tX}] \leq \frac{b}{b - a}e^{ta} - \frac{a}{b - a}e^{tb}.$$

Let $p = -a/(b - a) \in [0, 1]$ and $u = t(b - a)$. Then $b/(b - a) = 1 - p$. The right-hand side is $(1 - p)e^{-pu} + pe^{(1-p)u}$. Taking the logarithm and applying Taylor's theorem yields $\log(\mathbb{E}[e^{tX}]) \leq \frac{1}{8}t^2(b - a)^2$. $\square$

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## 3 Sub-Gaussian and Sub-Exponential Variables

### 3.1 Sub-Gaussian Random Variables

A random variable $X$ with $\mathbb{E}[X] = 0$ is **sub-Gaussian** with variance proxy $\sigma^2$ if:

$$\mathbb{E}[e^{tX}] \leq e^{\frac{\sigma^2 t^2}{2}} \quad \forall t \in \mathbb{R}.$$

**Equivalent characterizations:**

1. **Tail bound:** $\mathbb{P}(|X| \geq t) \leq 2e^{-t^2/(2\sigma^2)}$.
2. **Moment growth:** $(\mathbb{E}[|X|^p])^{1/p} \leq C\sigma\sqrt{p}$ for $p \geq 1$.
3. **Sub-Gaussian norm:** $\|X\|_{\psi_2} = \sup_{p \geq 1} p^{-1/2}(\mathbb{E}[|X|^p])^{1/p} < \infty$.

*Example.* Any bounded random variable $X \in [a, b]$ with $\mathbb{E}[X] = 0$ is sub-Gaussian with $\sigma = (b - a)/2$ (by Hoeffding's Lemma).

### 3.2 Sub-Exponential Random Variables

A random variable $X$ with $\mathbb{E}[X] = 0$ is **sub-exponential** with parameters $(\nu, \alpha)$ if:

$$\mathbb{E}[e^{tX}] \leq e^{\frac{\nu^2 t^2}{2}} \quad \text{for } |t| \leq \frac{1}{\alpha}.$$

**Equivalent characterizations:**

1. **Tail bound:** $\mathbb{P}(|X| \geq t) \leq 2e^{-t/\alpha}$ for $t \geq \nu^2/\alpha$.
2. **Sub-exponential norm:** $\|X\|_{\psi_1} = \sup_{p \geq 1} p^{-1}(\mathbb{E}[|X|^p])^{1/p} < \infty$.

*Example.* If $Z \sim \mathcal{N}(0,1)$, then $X = Z^2 - 1$ is sub-exponential.

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## 4 Concentration Inequalities for Sums

### 4.1 Hoeffding's Inequality

Let $X_1, \ldots, X_n$ be independent random variables such that $a_i \leq X_i \leq b_i$ almost surely. Let $\bar{X} = \frac{1}{n}\sum X_i$ and $\mu = \mathbb{E}[\bar{X}]$. Then for any $t > 0$:

$$\mathbb{P}(\bar{X} - \mu \geq t) \leq \exp\left(-\frac{2n^2 t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

*Proof.* Apply the Chernoff bound to $Z = \sum(X_i - \mathbb{E}[X_i])$. By independence and Hoeffding's Lemma,

$$\mathbb{E}[e^{sZ}] = \prod_{i=1}^n \mathbb{E}[e^{s(X_i - \mathbb{E}[X_i])}] \leq \prod_{i=1}^n e^{\frac{s^2(b_i-a_i)^2}{8}} = e^{\frac{s^2}{8}\sum(b_i-a_i)^2}.$$

Setting $s = 4nt/\sum(b_i - a_i)^2$ and applying the Chernoff bound yields the result. $\square$

### 4.2 Chernoff Bounds for Independent Bernoulli Variables

Let $X = \sum_{i=1}^n X_i$ where $X_i \sim \text{Bernoulli}(p_i)$ independently, and let $\mu = \mathbb{E}[X] = \sum p_i$. Then for any $\delta > 0$:

1. **Upper tail:** $\mathbb{P}(X \geq (1+\delta)\mu) \leq \exp\left(-\frac{\delta^2 \mu}{2+\delta}\right)$.
2. **Lower tail:** For $0 < \delta < 1$, $\mathbb{P}(X \leq (1-\delta)\mu) \leq \exp\left(-\frac{\delta^2 \mu}{2}\right)$.

### 4.3 Bernstein's Inequality

Let $X_1, \ldots, X_n$ be independent random variables with $\mathbb{E}[X_i] = 0$, $|X_i| \leq M$ almost surely, and $\text{Var}(X_i) = \sigma_i^2$. Let $\sigma^2 = \frac{1}{n}\sum \sigma_i^2$. Then for any $t > 0$:

$$\mathbb{P}\left(\frac{1}{n}\sum X_i \geq t\right) \leq \exp\left(-\frac{nt^2}{2\sigma^2 + 2Mt/3}\right).$$

### 4.4 McDiarmid's Inequality

Let $f : \mathcal{X}^n \to \mathbb{R}$ satisfy the bounded difference property: for each $i$, there exists $c_i > 0$ such that $|f(x) - f(x')| \leq c_i$ whenever $x$ and $x'$ differ only in the $i$-th coordinate. Then for any $t > 0$:

$$\mathbb{P}(f(X) - \mathbb{E}[f(X)] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

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## 5 Matrix Concentration

### 5.1 Matrix Chernoff Bound

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be independent, symmetric, positive semi-definite $d \times d$ matrices with $\mathbb{E}[\mathbf{X}_i] = \mathbf{0}$ and $\lambda_{\max}(\mathbf{X}_i) \leq R$ almost surely. Let $\mu_{\max} = \lambda_{\max}\left(\sum_{i=1}^n \mathbb{E}[\mathbf{X}_i^2]\right)$. Then for any $\delta \in (0, 1)$:

$$\mathbb{P}\left(\lambda_{\max}\left(\sum_{i=1}^n \mathbf{X}_i\right) \geq (1+\delta)\mu_{\max}\right) \leq d \cdot \exp\left(-\frac{\delta^2 \mu_{\max}}{3R}\right).$$

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## 6 Application: Discrepancy and the Probabilistic Method

Let $V$ be a set of $n$ elements and $\mathcal{F}$ a family of subsets. A coloring $\chi : V \to \{-1, 1\}$ is chosen uniformly at random. For a set $S \in \mathcal{F}$, the discrepancy is $d(S) = \sum_{v \in S} \chi(v)$.

Since $\chi(v)$ are independent Rademacher variables, $d(S)$ is sub-Gaussian:

$$\mathbb{P}(|d(S)| \geq t) \leq 2\exp\left(-\frac{t^2}{2|S|}\right) \leq 2\exp\left(-\frac{t^2}{2n}\right).$$

By the union bound over $m = |\mathcal{F}|$ sets:

$$\mathbb{P}\left(\max_{S \in \mathcal{F}} |d(S)| \geq t\right) \leq 2m\exp\left(-\frac{t^2}{2n}\right).$$

Setting the right-hand side to $1/2$ (to prove existence via the probabilistic method) yields $t = O(\sqrt{n \log m})$. This establishes that any set system can be colored with discrepancy at most $O(\sqrt{n \log m})$, a foundational result in discrepancy theory.